

Likewise - the 2025 national run

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Likewise — the 2025 national run

What was run. The Likewise engine, specification v1.5.0, over the FFIEC 2025 Snapshot National Loan Level Dataset: 13,543,606 records from 4,782 filers. One scan per filer-year — the national file is never scanned as a unit, because the question the product asks is about a lender's own filing against its own filing.

Date. 2026-09-10 · **Author.** Arun Kadambi · **Specification.** materiality 1.5.0, budget 1.1.0, reasons 1.0.0 · **Publication snapshot.** ffiec_flg_2025_2026-09-15

Filers attempted: 2,883 — every filer in the country with at least one denial citing a reason the public record can carry and at least one approval to compare it against. 2,837 scanned cleanly, 42 refused by the post-treatment guard, 4 refused at load.

No lender is named anywhere in this report. Identifiers are the engine's egress masks.

1. The headline, stated plainly

Across the whole country the engine produced **372 findings** from **2,013,872 denials**. That is not a disappointing yield. It is the result, and the direction it points in is the opposite of an alarm.

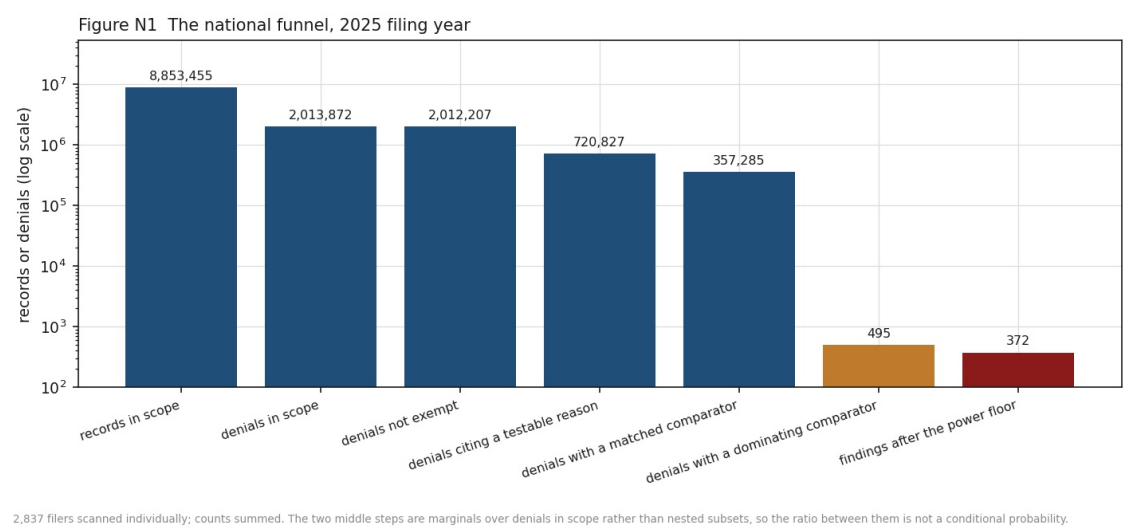
Denied applicants sit WORSE than chance on the dimension their own lender's stated reason names. The filer-level pooled test is signed so that a positive z means denials sit *better* than chance on their cited dimension — the direction that would make a stated reason hard to justify. Nationally:

- median filer-level $z = -1.1785$
- 81% of testable filers are negative
- the largest filers are the most negative: the three biggest score $z = -64.5, -77.4$ and -11.4
- combining the 1,270 filer-level tests by Fisher's method gives $\chi^2 = 2,325$ on 2,540 degrees of freedom — **below** its null mean of 2,540, with the *lower* tail at $p = 0.0010$

Read carefully, that says the stated reasons in the 2025 national record are **more consistent with the public data than chance would produce**, systematically and at scale.

Two honest qualifications. First, this is what the public fields can see: a lender denying for debt-to-income does file applicants with worse debt-to-income, and nothing here establishes that the denial was *caused* by it. Second, the direction is partly mechanical — a reason is cited because the underlying quantity was bad, and the published record carries a coarsened version of that quantity. The result is a strong negative control on the method, not a clean bill of health for the industry.

2. The funnel



step	count	share of previous
records in scope	8,853,455	—
denials in scope	2,013,872	22.7%
denials not exempt	2,012,207	99.9%
denials citing a testable reason	720,827	35.8%
denials with a matched comparator	357,285	49.6%
denials with a dominating comparator	495	0.1%
findings after the power floor	372	75.2%

The two middle steps are MARGINALS over denials in scope, not nested subsets: the engine computes testability without conditioning on matching and matching without conditioning on testability. Nationally the counts happen to nest; at filer level they do not, and the ratio between them is not a conditional probability.

Three of these steps deserve their own sentence.

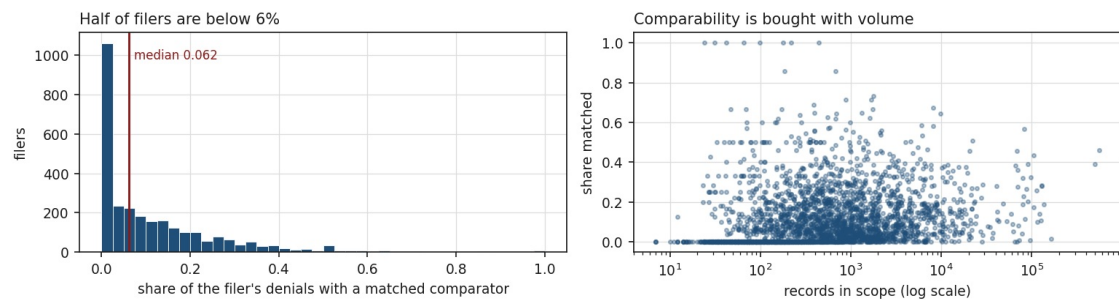
230,237 denials cite only reasons the public record cannot carry — credit history, employment history, unverifiable information, incomplete application. The file has no column for any of them. A product that quietly counted these as “nothing found” would be reporting the regulator’s field list as a clean bill of health.

73,243 denials are testable in principle and unresolvable in practice, because the published values are too coarse to separate the pair. That is the resolution budget refusing to manufacture a difference the file cannot express.

13,687 findings were suppressed by k-anonymity before publication, which is more than thirty times the number that survived. Disclosure control, not statistics, is what removes most of what the engine finds.

3. What the design can see, by filer size

Figure N2 Matching is the binding constraint, and it scales with size

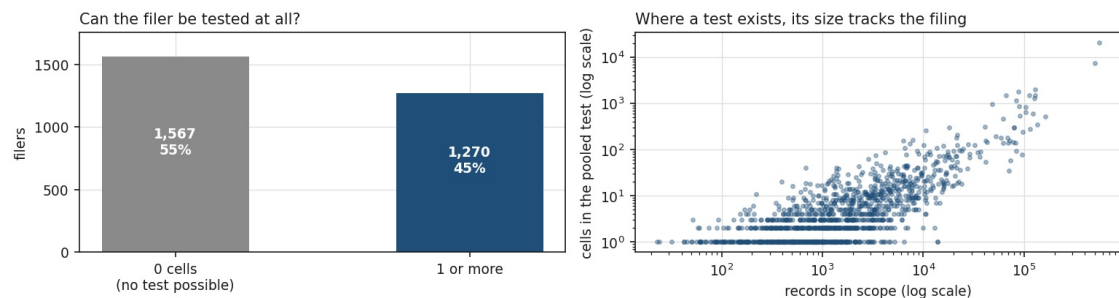


A denial with no matched comparator is not a clean denial. It is one the public record cannot speak to, and the engine reports it as unaddressed.

filing size (records in scope)	filers	denials	median share of denials with a comparator	filers with any testable cell	findings
under 100	513	4,881	0.0%	7%	0
100 - 999	1,391	83,345	6.2%	33%	0
1,000 - 9,999	793	392,295	10.2%	80%	6
10,000 or more	140	1,533,351	13.7%	96%	366

The median filer gets a comparator for **6.2%** of its denials. The mean is 11.3%, pulled up by the large filers, and the tenth percentile is zero.

Figure N3 Half the country's filers contribute no cell to any test

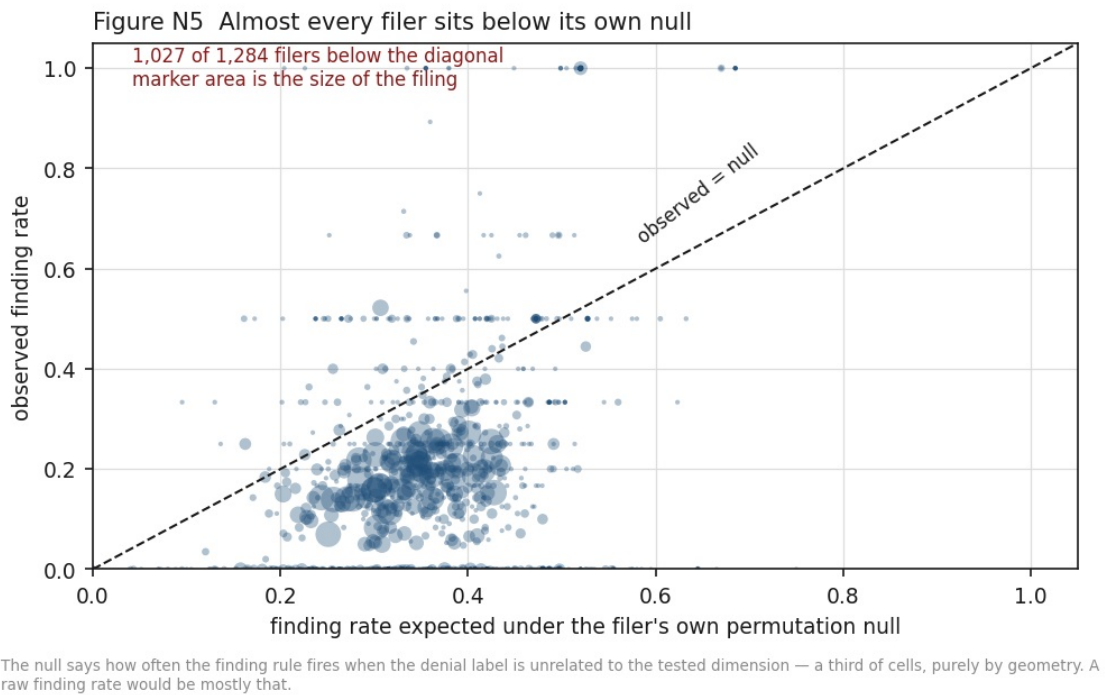


A cell needs a denial and at least one approved application matched to it on the exact key and inside both bands. The median filer has none.

1,567 of 2,837 filers (55%) contribute no cell to any test at all. For those lenders the product's honest output is that the public record cannot answer the question — which is a real answer, and one they cannot get anywhere else, but it is not the answer the product was built to give.

This is the same finding the simulation study reached from the other direction: the binding constraint is not statistical power, it is whether a comparison exists.

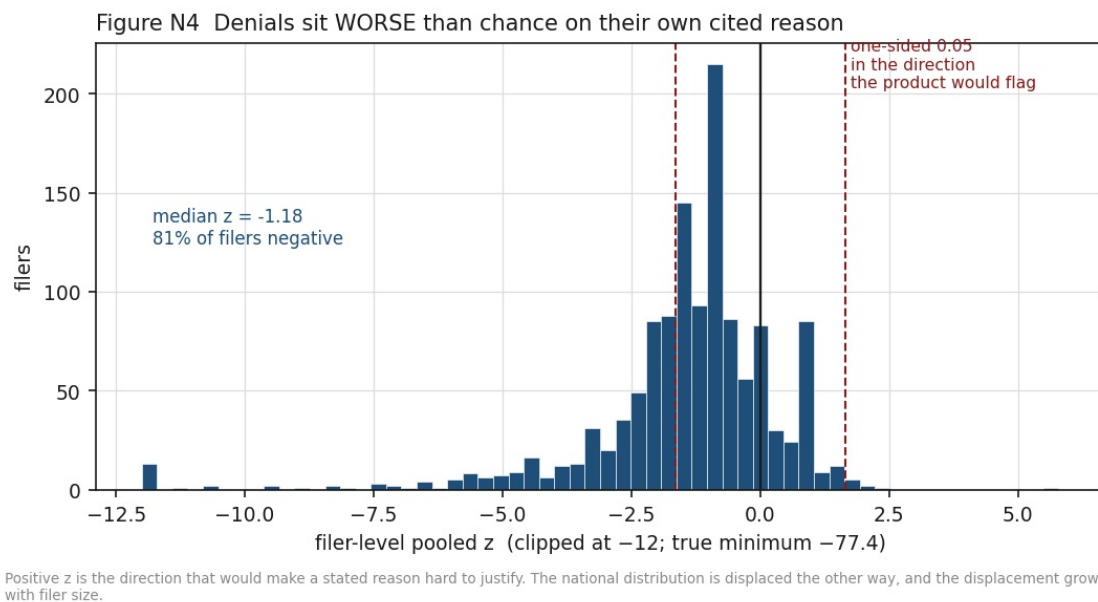
4. Against each filer's own null



The permutation null asks how often the finding rule would fire if the denial label were unrelated to the tested dimension. The answer is: often. Across filers the null mean clusters between 0.2 and 0.5 — a small cell with a threshold below its spread produces a finding by geometry alone.

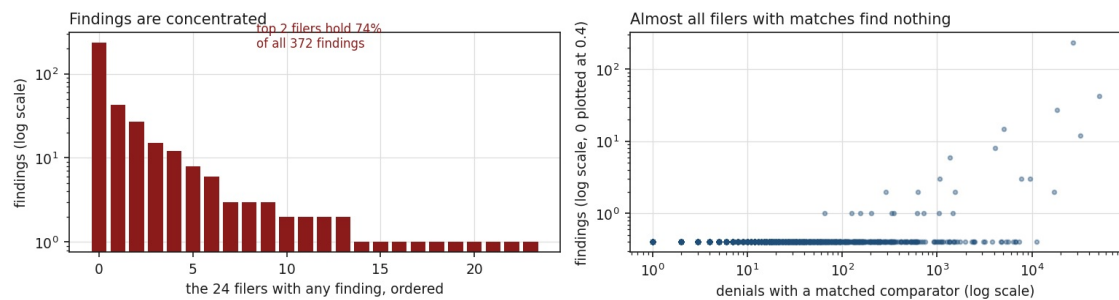
Against that baseline, of the filers where a null could be computed: **166 sit below their own null**, 1,115 sit inside it, and **3 sit above it**.

Three. In the whole country.



5. Where the findings are

Figure N7 372 findings out of 357,285 matched denials



A finding means the public record does not by itself support the stated reason. It is a queue item for human review, never a determination.

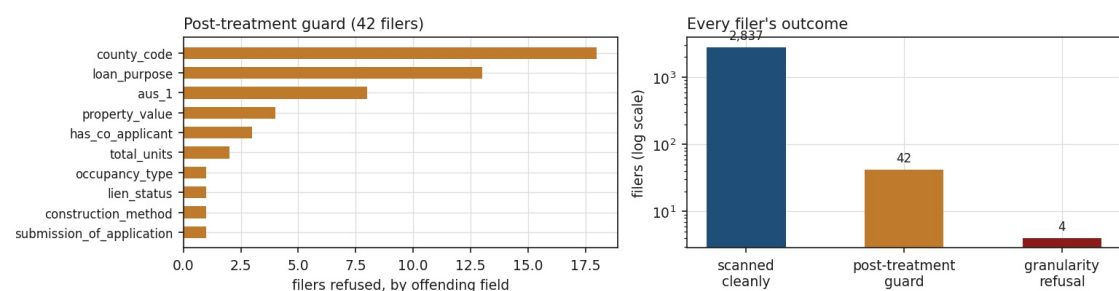
Only **24 filers** produced any finding at all, and the two largest hold most of them. The highest finding rate among any filer's matched denials is 1.54%.

filer (masked)	records	denials	with a comparator	dominating	findings	pooled z
LEI-35d99003eb	500,753	69,354	26,982	254	234	-64.4864
LEI-e45fc829f8	554,837	110,549	50,828	49	43	-77.4283
LEI-74052bf471	91,166	46,502	18,239	66	27	-17.4366
LEI-c5924d2f07	68,026	11,093	5,085	20	15	-4.3519
LEI-fe0b5505d1	104,367	74,949	32,626	37	12	-4.7867
LEI-ff9217ca46	112,588	15,894	4,127	8	8	-16.8343
LEI-9e28f7353a	79,744	7,301	1,366	7	6	-8.3104
LEI-cc2c206532	77,862	24,563	1,064	3	3	-5.2945
LEI-d3748e8174	83,233	13,664	7,751	3	3	-6.2726
LEI-110ff84e25	128,056	34,164	9,638	4	3	-14.7964

A finding is a queue item, not a determination. It says the public record does not by itself support the stated reason for that denial — no more than that.

6. Refusals, and one defect the run found

Figure N6 Refusals are results, and they are rare



The four granularity refusals cost roughly 190,000 records over six values, every one of them exactly 2,147,483,647 — the largest 32-bit signed integer, not a property value.

The post-treatment guard refused 42 filers (1.5%). The guard detects an exact-match field whose value set is disjoint between denials and approvals, and refuses rather than reporting “no comparable files” for a reason that has nothing to do with lending. The offending fields, in order: county_code (18), loan_purpose (13), aus_1 (8), property_value (4), has_co_applicant (3).

Most of these are small filers where two or three records separate by chance. The guard cannot distinguish that from a field genuinely determined by the outcome, and at small n it will not be able to. Worth stating as a known limit rather than presenting the count as evidence of systematic post-treatment coding.

Four filers were refused at load — and this one is a defect, not a design choice.

records in the filing	field	conforming	of	fraction
53,631	property_value	58,362	58,363	0.999983
105,748	property_value	129,319	129,321	0.999985
1,397	property_value	1,398	1,399	0.999285
263	property_value	221	223	0.991031

Six records in the entire national file carry a property_value that is not on the \$10,000 bin-midpoint grid, and **every one of them is exactly 2,147,483,647** — the largest signed 32-bit integer. That is an overflow artefact in the published file, not un-coarsened data. The loader’s rule is every value or none, so those six records refused four whole filings, two of them large: roughly 190,000 records lost to six values.

The rule is right and the diagnosis is wrong. 2147483647 is not in the schema’s list of HMDA numeric sentinels (1111, 8888, 9999, -1), so it survives normalisation as a number and then fails the granularity check, which reports it as “data is not at publication granularity” and advises loading the file as the regulator publishes it — which is exactly what was done. **Recommended fix: add 2147483647 to the numeric sentinels**, on the evidence of these six records, so it normalises to absent like every other sentinel. That is a schema claim about the regulator’s publication and it should be evidenced in the specification rather than patched in code.

I did not make that change during this run. You asked for the model as written, and changing the schema mid-run would have made the results a mixture of two standards.

7. What this run does not establish

It is not a finding about any lender. Every number here is aggregate, every identifier is masked, and the product’s own output is a review queue.

The direction is not proof of good behaviour. A negative pooled z says the stated reasons line up with the public fields better than chance. It cannot see credit score, reserves, documentation quality or anything else absent from the file, and the Rosenbaum sensitivity bound on findings of this size tolerates very little unmeasured confounding.

Comparability is conditional on volume. The filers the product can say most about are the largest, and the filers it was designed to help are the smallest. Figure N2 is the shape of that problem.

Nothing was pre-registered. Every scan carries analysis_status: exploratory, because no preregistration digest was supplied. Under the product’s own governance these results may not be published as confirmatory, and this report is not one.

The changes the earlier national review recommended were not applied. Blocking on MSA-or-state instead of county, ordinal comparison over the published DTI bands, and one-sided control tolerances were all measured to increase what the design can see, by roughly six-fold in the powered fraction. This run is the baseline they have to beat.

8. Reproducing it

```
python cloudrun.py --shard i --of n      # one scan per filer, resumable
python natreport.py                      # aggregate to results/national.json
python natfigures.py                    # the seven figures
python natreport_md.py                  # this report
```

Every scan id is `sha256(lei | year | spec version | publication snapshot | curated snapshot)`, so re-running an identical tuple overwrites the same record rather than accumulating near-duplicates, and a result computed on one machine is the same result on another.

One engine change was required before any of this could run. The Snapshot National Loan Level Dataset is a third CSV layout. The loader detected layouts by marker column, and the Snapshot's marker (`combined_loan_to_value_ratio`) was the template's — while the Snapshot publishes income in thousands, as the Data Browser does, and the template path does not scale it. Every income would have come out a thousand times too small with nothing about the load looking wrong. It failed safe only because the Snapshot lacks two derived columns that the template requires. Detection now scores each layout against its full required column set and refuses ambiguity; the regression test is `test_the_snapshot_layout_is_not_read_as_a_template`.

Scale of the run. 2,553,576 candidate pairs, 2,476,818 matched pairs, 2837 scans, median 2.5 s per scan and 189 s for the largest filer.